NeuroTIMM: Harmonizing the TIMM Model Zoo with Human Visual Strategies

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Abstract

In previous work by (Fel et al., 2022), they showed that "harmonizing" deep neural networks with human psychophysics to force them to rely on the same visual features as human observers aligned models with human perception while maintaining classification accuracy (Fel et al., 2022; Linsley et al., 2023b). Here, we scale up this approach with ClickMe 2.0, a larger and more robust behavioral dataset, and more efficient harmonization methods that make it feasible to align models across the full PyTorch Image Models (TIMM) library. Harmonized models exhibit substantially improved alignment with human visual strategies on ImageNet, which also yields higher correspondence with electrophysiological recordings from macaque visual cortex (Arcaro & Livingstone, 2017), and improved robustness against visual perturbations. We release all the harmonized weights as an open resource for the community.

Introduction

Pretrained deep neural networks (DNNs), and in particular models from the PyTorch Image Models library (TIMM; Wightman, 2019), have become a standard tool not only in computer vision but also in computational neuroscience, where they serve as the leading models of the primate ventral visual stream (Schrimpf et al., 2018; Serre, 2019; Yamins et al., 2014). However, others have recently shown that as DNNs have scaled in size and benchmark accuracy, they have become *worse*—not better—models of biological vision (Linsley et al., 2023b, 2025). Today's most accurate models rely on visual features that are increasingly misaligned with those used by human and non-human primate observers. Given that these models are widely used as proxies for biological vision, it is important to develop methods to realign them with human and primate perception.

The *neural harmonizer* is a training routine that co-optimizes DNNs for object classification and for alignment with human feature importance maps collected through the ClickMe game (Linsley et al., 2017, 2019). Harmonizing a small set of architectures significantly improved their alignment with human visual strategies, their ability to predict neural responses in primate inferotemporal cortex, and their robustness to adversarial perturbations—while preserving competitive classification accuracy relative to standard training routines(Fel et al., 2022; Linsley et al., 2023a, 2023b).

Here we scale up this approach with ClickMe 2.0, a larger and more robust behavioral dataset, and harmonization methods efficient enough to cover the full TIMM library. We assess harmonized models on alignment with human visual strategies, neural predictivity against macaque visual cortex recordings (Arcaro & Livingstone, 2017), and robustness to visual perturbations. All harmonized weights are released as an open resource.

Methods

ClickMe 2.0

ClickMe 2.0 is a redesigned version of the ClickMe game that collects human feature importance maps at a greater scale and with improved reliability. As in the original game (Linsley et al., 2017, 2019), participants highlight the image regions they rely on to recognize objects. The new version features a streamlined web interface and incorporates stricter quality-control filters to reduce noise. The resulting dataset is substantially larger than its predecessor and provides denser coverage of the ImageNet label space.

Harmonization at scale

The neural harmonizer fine-tunes a pretrained DNN by jointly minimizing a classification loss and a perceptual alignment loss that encourages the model's gradient-based saliency maps to match ClickMe human feature importance maps (Fel et al., 2022). The original harmonization work harmonized models from scratch, which is computationally expensive and limited to a few architectures. Here we adopt a fine-tuning approach: we start from pretrained TIMM weights and harmonize with a reduced number of epochs, making it practical to scale across the library. we fine-tune pretrained weights for just 30 epochs rather than training from scratch. We optimize using AdamW (with model-specific learning rates) under a cosine annealing schedule, applying a multi-scale focal cross-entropy loss between standardized saliency and ClickMe distributions. The alignment loss weight is cosine-ramped from 0 to 1 over training, allowing the model to consolidate classification performance before progressively increasing alignment.

In submission to the *9th Conference on Cognitive Computational Neuroscience* (CCN 2026).

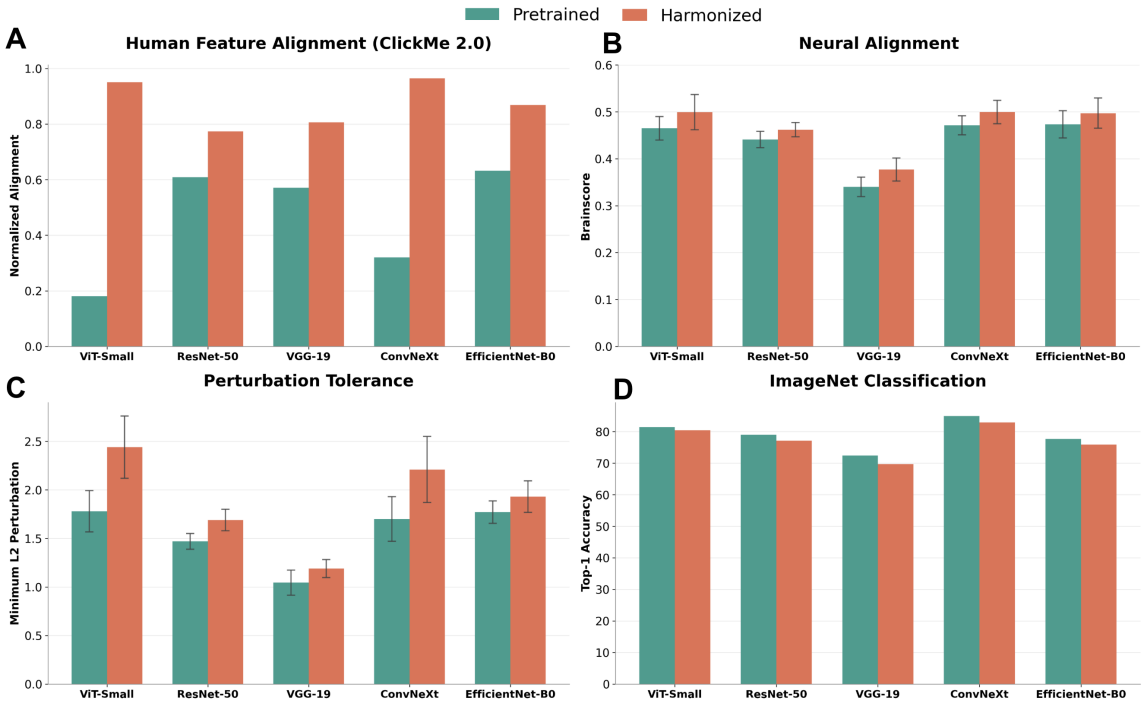

Figure 1: TIMM Pretrained Models vs. Harmonized Models. A) Human Feature Alignment on ClickMe 2.0. B) Neural Alignment(Arcaro & Livingstone, 2017). C) Perturbation Tolerance to L2-PGD Attacks. D) ImageNet Top-1 Accuracy.

Results

For all metrics, we compare each harmonized model against its unharmonized TIMM baseline. We evaluate harmonized models along three axes:

1. **Human alignment.** We measure the Spearman correlation between model saliency maps and human feature importance maps on a held-out set of ImageNet validation images annotated with ClickMe 2.0, normalized by the human-to-human ceiling of 0.73. Standard TIMM models exhibit poor alignment with human visual strategies, particularly at larger scales: `ViT-Small` and `ConvNeXt` score just 0.18 and 0.32, respectively, on ClickMe 2.0 (Fig. **??**D). Harmonization produces large gains across all architectures, with the most misaligned models improving the most—`ViT-Small` alignment rises to 0.95, `ConvNeXt` to 0.88.

2. **Neural predictivity.** We assess how well model representations predict electrophysiological recordings from macaque visual cortex using the Arcaro dataset (Arcaro & Livingstone, 2017). Specifically, we map spatially-aligned layer activations to neural responses using Partial Least Squares regression, evaluated via leave-one-out cross-validation. Harmonized models outperform TIMM baselines on the Arcaro dataset (Fig. A) for both CNNs and transformers. For example, brainscore for harmonized `ViT-Small` rises from 0.46 to 0.50. Co-optimizing for human visual strategies yields representations closer to activity measured in primate visual cortex.

3. **Robustness.** We measure classification accuracy under standard visual perturbations and adversarial attacks, quantified as the minimum L2-perturbation required to cause misclassification. Harmonized models are more robust to adversarial perturbations (Fig. C). For instance, `ConvNeXt` perturbation tolerance under L2-PGD attacks improves from 1.7 to 2.5. By relying on human-like features rather than dataset-specific shortcuts, harmonized models become substantially more resilient to visual distortions.

Accuracy trade-off. Harmonization produces a small decrease in ImageNet top-1 accuracy (e.g., `ResNet-50` drops from 81.40% to 80.37%; Fig. B). This is expected: the models abandon distributed features in favor of a more robust, biologically grounded local feature space.

Discussion

These results show that harmonization scales effectively across a diverse set of architectures when combined with ClickMe 2.0 and efficient fine-tuning. The generalization from behavioral alignment to neural predictivity in macaque cortex reinforces the view that human perceptual strategies reflect something fundamental about biological visual processing (Linsley et al., 2025).

References

Arcaro, M. J., & Livingstone, M. S. (2017). A hierarchical, retinotopic proto-organization of the primate visual system at birth. *eLife*, *6*, e26196. https://doi.org/10.7554/eLife.26196.

Fel, T., Rodriguez, I. F., Linsley, D., & Serre, T. (2022). Harmonizing the object recognition strategies of deep neural networks with humans. *Advances in Neural Information Processing Systems (NeurIPS)*.

Linsley, D., Eberhardt, S., Sharma, T., Gupta, P., & Serre, T. (2017). What are the visual features underlying human versus machine vision? *IEEE International Conference on Computer Vision Workshops (ICCVW)*, 2706–2714.

Linsley, D., Feng, P., Boissin, T., Ashok, A. K., Fel, T., Olaiya, S., & Serre, T. (2023a). Adversarial alignment: Breaking the trade-off between the strength of an attack and its relevance to human perception. *arXiv preprint*.

Linsley, D., Feng, P., & Serre, T. (2025). Better artificial intelligence does not mean better models of biology [In press].

Linsley, D., Rodriguez, I. F., Fel, T., Arcaro, M., Sharma, S., Livingstone, M., & Serre, T. (2023b). Performance-optimized deep neural networks are evolving into worse models of inferotemporal visual cortex. *Advances in Neural Information Processing Systems (NeurIPS)*.

Linsley, D., Shiebler, D., Eberhardt, S., & Serre, T. (2019). Learning what and where to attend. *International Conference on Learning Representations (ICLR)*.

Schrimpf, M., Kubilius, J., Hong, H., Majaj, N. J., Rajalingham, R., Issa, E. B., Kar, K., Bashivan, P., Prescott-Roy, J., Geiger, F., Schmidt, K., Yamins, D. L. K., & DiCarlo, J. J. (2018). Brain-score: Which artificial neural network for object recognition is most brain-like? *bioRxiv*. https://doi.org/10.1101/407007.

Serre, T. (2019). Deep learning: The good, the bad, and the ugly. *Annual Review of Vision Science*, *5*, 399–426. https://doi.org/10.1146/annurev-vision-091718-014951.

Wightman, R. (2019). Pytorch image models. https://doi.org/10.5281/zenodo.4414861.

Yamins, D. L. K., Hong, H., Cadieu, C. F., Solomon, E. A., Seibert, D., & DiCarlo, J. J. (2014). Performance-optimized hierarchical models predict neural responses in higher visual cortex. *Proceedings of the National Academy of Sciences*, *111*(23), 8619–8624. https://doi.org/10.1073/pnas.1403112111.